SEKER AI

AI • DATA • SOFTWARE • AUTOMATION

AI & MACHINE LEARNING INTERNSHIP | PROJECT 06

AI-Powered Demand Forecasting

& Predictive Analytics System

An end-to-end machine learning project in which interns analyse historical business data and predict future demand using time-series analysis, feature engineering, forecasting models and interactive analytics — then convert those predictions into decisions a real business can act on.

DOCUMENT	**Project Assignment Brief**	TRACK	**AI & Machine Learning**
PROJECT	**Project 06**	LEVEL	**Intermediate — Advanced**
ISSUED BY	**seKer AI**	SUGGESTED EFFORT	**2 – 3 weeks**
VERSION	**1.0**	DELIVERABLE	**Repository, dashboard, demo**

ASSIGNMENT BRIEF

What this project asks you to build

Forecasting is the quiet engine behind almost every planning decision a business makes. How much stock to hold, how many staff to roster, how much raw material to purchase, when to discount and when to hold price — each is a bet on future demand. When the bet is informed, capital works efficiently. When it is not, the business pays twice: once for inventory that does not sell, and again for sales it could not fulfil.

In Project 06 you will build a complete demand forecasting system: from a raw historical dataset, through cleaning and exploratory analysis, into engineered time-series features, competing forecasting models, honest evaluation, and finally an interactive dashboard that turns numbers into decisions. The technical work is only half the assignment. The other half is the recommendation you hand to the business at the end.

> **HOW TO READ THIS DOCUMENT**
>
> Sections 01–03 frame the problem and objectives. Sections 04–14 are the technical build, in the order you should work through them. Sections 15–19 turn results into business value. Sections 20–26 define exactly what you submit and how it will be assessed. Work sequentially — each stage depends on the quality of the one before it.

DOCUMENT MAP

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01 PROJECT OVERVIEW

You will develop an AI-powered forecasting system capable of learning from historical demand data and projecting demand forward over a defined horizon. The scenario should mirror a real business context — retail sales, product demand, inventory planning, order forecasting or resource planning — and the output must be usable by someone who will never read your code.

END-TO-END SYSTEM PIPELINE



Where a demand forecast is actually used

BUSINESS FUNCTION	DECISION THE FORECAST INFORMS
Inventory	How much stock to hold per product and location, and at what reorder point
Purchasing	When to place orders with suppliers and in what quantity, given lead times
Production	How much to manufacture in each cycle to avoid idle capacity or backlog
Staffing	How many people to roster for expected volume, shift by shift
Supply chain	How to allocate stock across warehouses, stores and delivery routes
Marketing	When to promote, which products need demand stimulus, which do not
Sales planning	Realistic targets, quota setting and revenue projection
Resource allocation	Where to place budget, equipment and capacity for the next period

What poor forecasting costs

FAILURE MODE	CONSEQUENCE FOR THE BUSINESS
Overstocking	Working capital frozen in unsold goods; storage cost; markdowns and write-offs
Stockouts	Unfulfilled orders, emergency purchasing at premium prices, disrupted operations
Lost sales	Revenue that never arrives, plus customers who switch to a competitor and stay there
Higher operating cost	Overtime, expedited freight, rush production and last-minute correction
Poor resource use	People and equipment idle in quiet periods and overwhelmed in peaks

> **THE STANDARD FOR THIS PROJECT**
>
> A forecast that is merely produced is not a deliverable. A forecast becomes a deliverable when it is validated honestly, communicated clearly, and paired with a specific recommendation the business can act on this week.

02 BUSINESS PROBLEM

Consider a retail business selling hundreds or thousands of distinct products across one or more locations. Every transaction has been recorded for several years, so the company is not short of data — it is short of foresight. Planning meetings still rely on last year's figures, a spreadsheet and the instinct of whoever has been there longest.

Data the business already holds

FIELD	DESCRIPTION	WHY IT MATTERS TO THE FORECAST
Date	Transaction or aggregation date	The backbone of every time-series feature
Product / SKU	Item identifier and category	Enables per-product and per-category forecasting
Quantity	Units sold or ordered	Usual target variable
Price	Unit price at time of sale	Captures elasticity and discount effects
Promotion	Campaign or discount flag	Explains spikes that would otherwise look like noise
Store / location	Branch, region or warehouse	Demand patterns differ sharply by location
Holiday	Public or seasonal holiday flag	Explains recurring calendar-driven surges
Season	Weather or trading season	Drives medium-term cyclical behaviour

The questions management wants answered

- What will demand look like next week, and how confident should we be in that number?
- Which products will see higher demand next month, and by roughly how much?
- Which products are quietly declining and should be de-stocked before they become dead inventory?
- When should inventory be replenished, given supplier lead times?
- Which seasonal patterns should the business be preparing for right now?

> **OBJECTIVE**
> Build a predictive system that answers these questions from historical data — with quantified accuracy, visible uncertainty, and recommendations expressed in the language of the business rather than the language of the model.

03 PROJECT OBJECTIVES

Fourteen objectives define the scope of Project 06. Each maps to a phase of the delivery pipeline and each produces evidence you will be assessed on.

#	OBJECTIVE	PHASE
01	Understand the forecasting dataset and its business context	Foundation
02	Perform comprehensive data cleaning and validation	Foundation
03	Conduct thorough exploratory data analysis	Analysis
04	Identify long-term trend and repeating seasonality	Analysis
05	Detect unusual demand patterns, spikes and outliers	Analysis

#	OBJECTIVE	PHASE
06	Engineer meaningful time-series features	Modelling
07	Establish a baseline forecasting model	Modelling
08	Experiment with multiple forecasting approaches	Modelling
09	Compare model performance on a fair, chronological split	Modelling
10	Generate future demand predictions over a defined horizon	Modelling
11	Evaluate forecasting accuracy with appropriate metrics	Modelling
12	Visualise historical demand against predicted demand	Delivery
13	Build an interactive forecasting dashboard	Delivery
14	Generate actionable business recommendations	Delivery

04 DATASET REQUIREMENTS

Select a publicly available and legally usable dataset that presents a genuine forecasting problem. Suitable domains include retail sales, product demand, store sales, e-commerce orders, inventory demand, transportation demand, energy consumption and restaurant demand. You may choose a different dataset entirely, provided it contains a meaningful time-series signal and enough history to validate against.

Preferred dataset characteristics

ATTRIBUTE	REQUIREMENT	NOTE
Date field	Required	Continuous, unambiguous, and at a consistent granularity
Product or item identifier	Required	Even a single product is acceptable if history is long
Sales quantity	Required	The default target variable
Revenue	Preferred	Allows value-weighted as well as unit-based insight
Store or location	Where available	Unlocks regional and per-store analysis
Price	Where available	Supports price–demand and elasticity analysis
Promotion flag	Where available	Explains otherwise inexplicable demand spikes
Holiday / season	Where available	Can be derived from the date if not supplied
History length	Required	At least two full seasonal cycles, ideally more

Documentation you must provide

- Dataset source, licence and the date you downloaded it
- Number of records and number of features
- Time period covered and the granularity of observations
- Target variable and its unit of measurement
- Important limitations — gaps, known collection issues, definition changes, coverage bias

DATA LICENSING

Confirm that the dataset licence permits your intended use before you build on it. Record the licence in your README. Never publish proprietary, confidential or personally identifiable data to a public repository.

05 DATA UNDERSTANDING & CLEANING

Before any modelling, carry out a complete initial investigation. The purpose is not to tidy the data but to understand what it is, where it is weak, and which of its irregularities are real.

Initial investigation

CHECK	WHAT YOU ARE LOOKING FOR
Dataset dimensions	Row and column counts; whether the volume matches expectations
Data types	Dates parsed as dates, numerics as numerics, identifiers as categories
Missing values	Which columns, how much, and whether the gaps cluster in time
Duplicate records	Exact duplicates and repeated keys such as date–product–store
Unique products / stores	Cardinality, and how much history each entity actually has
Date range	Start, end, granularity and any missing calendar dates
Target distribution	Shape, skew, proportion of zeros, presence of long tails
Statistical summary	Mean, median, quartiles, standard deviation, min and max

Cleaning operations

OPERATION	GUIDANCE
Missing-value handling	Distinguish a missing record from a genuine zero-demand day; choose forward fill, interpolation or explicit zero deliberately and justify it
Duplicate removal	Remove exact duplicates; investigate rather than delete repeated keys
Date conversion	Parse to datetime, set as index, reindex to a complete calendar
Incorrect values	Negative quantities, impossible prices, dates outside the stated range
Outlier investigation	Flag and explain before deciding; never silently drop
Aggregation	Roll up to the granularity your business question requires — daily, weekly or monthly

DO NOT BLINDLY REMOVE UNUSUAL OBSERVATIONS

A demand spike is often the most valuable signal in the dataset. Promotions, holidays, weather events and product launches all produce legitimate extremes. Investigate every outlier, label the ones you can explain, and only remove a point when you can demonstrate it is a recording error.

06 EXPLORATORY DATA ANALYSIS

Time-series EDA has a different objective from ordinary EDA: you are not only describing the data, you are diagnosing its structure so that you can choose an appropriate model. Work through trend, seasonality, distribution, product behaviour and location behaviour in turn.

Trend

Identify long-term direction. Is demand structurally growing, flat or declining? Rolling means over 30, 60 and 90 days will reveal movement that daily data hides. Note any level shifts — points where the series steps up or down and stays there — because they often mark a business change the model cannot infer on its own.

Seasonality

Look for repeating structure at daily, weekly, monthly and yearly frequencies. Weekly seasonality is almost universal in retail; annual seasonality needs at least two complete years to be trustworthy. Distinguish seasonality (fixed period) from cyclicality (variable period), because only the former can be encoded reliably as a calendar feature.

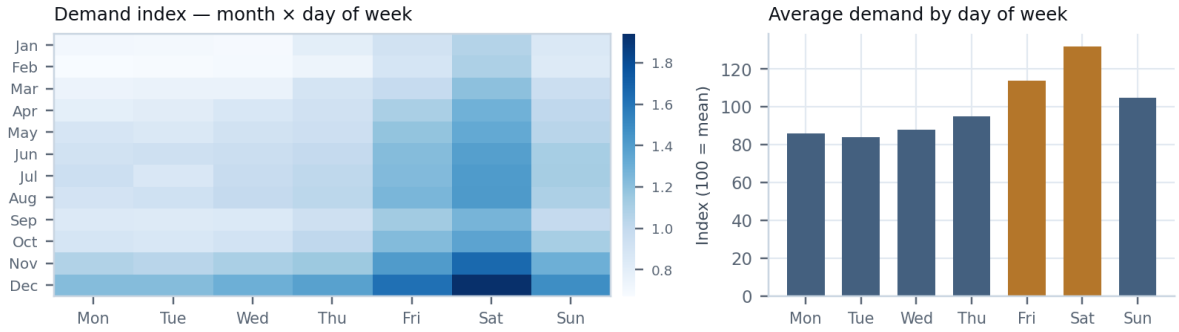


Figure 1 — Example seasonality view: month × day-of-week demand index and average demand by weekday. Weekend and December peaks are visible before any model is trained.

Demand distribution

Report average, minimum and maximum demand, variability, and the frequency and magnitude of spikes. A right-skewed distribution with many low values and occasional large ones is common and has direct consequences for both metric choice and model choice.

Product-level analysis

Where the dataset supports it, compare high-demand against low-demand products and fast-moving against slow-moving items. Intermittent products with many zero-demand days need different treatment from steady sellers, and it is better to discover that now than after training.

Store or location analysis

Where location data exists, compare demand levels, growth rates and seasonal shape across stores or regions. Two stores with identical average sales can behave very differently week to week.

Visualisations to produce

CHART	QUESTION IT ANSWERS
Time-series line chart	What has demand actually done over the full history?
Rolling average overlay	What is the underlying trend once noise is removed?
Seasonal / subseries plot	Which periods repeat, and how strong is the effect?

CHART	QUESTION IT ANSWERS
Bar chart by category	Which products, categories or stores dominate volume?
Heatmap	How do two calendar dimensions interact, e.g. month × weekday?
Distribution plot	How is demand spread, and how heavy is the tail?
Product comparison	Which items are growing and which are fading?

> **EVERY CHART NEEDS A SENTENCE**
> A visualisation without interpretation is decoration. Under each important figure, write one or two lines stating what the chart shows and what it implies for the forecasting approach. This is assessed.

07 TIME-SERIES DECOMPOSITION

Decomposition separates an observed series into components that can be reasoned about independently. It is the fastest way to understand what a model will need to learn.

COMPONENT	WHAT IT REPRESENTS	HOW IT GUIDES MODELLING
Trend	Long-term movement in the level of demand	Determines whether differencing or a trend term is needed
Seasonality	Repeating pattern at a fixed period	Sets the seasonal period for SARIMA and the calendar features for ML models
Residual / noise	What remains once trend and season are removed	Structure left here means information is still unmodelled

Techniques

- Moving averages — the simplest way to expose trend and suppress high-frequency noise
- Seasonal decomposition (additive or multiplicative) — choose multiplicative when seasonal swing grows with the level of demand
- Rolling statistics — rolling mean and rolling standard deviation to inspect stability of level and variance over time
- Stationarity checks — where you intend to use ARIMA-family models, test and document the differencing applied

Report which decomposition you used, why, and what it told you. If the residuals still contain visible structure, say so — that observation directly justifies the features you build next.

08 FEATURE ENGINEERING

Feature engineering is where most of the achievable accuracy in machine-learning forecasting is won. Each feature should encode a hypothesis about demand, and you should be able to state that hypothesis in one sentence.

GROUP	FEATURE	PURPOSE
Lag	lag_1, lag_7, lag_14, lag_28	Most recent demand and same-weekday demand from previous cycles; the strongest predictors in most retail series
Rolling	rolling mean 7 / 14 / 28	Smoothed recent level, resistant to single-day noise

GROUP	FEATURE	PURPOSE
Rolling	rolling standard deviation	Recent volatility — tells the model how reliable the recent level is
Calendar	day of week, day of month	Captures weekly rhythm and pay-cycle effects
Calendar	month, quarter, year	Captures annual seasonality and long-run drift
Calendar	weekend flag, holiday flag	Isolates known non-trading-day behaviour
Business	price, discount, promotion	Explains demand movements caused by commercial decisions
Business	store, product category	Allows one model to learn shared structure across entities

LEAKAGE WARNING

Every lag and rolling feature must be computed using only information available at prediction time. A rolling mean centred on the current row, or a lag built after shuffling, will produce excellent validation scores and a worthless model. Build features on a strictly sorted series and verify one row by hand.

09 TRAIN / VALIDATION / TEST STRATEGY

A random train-test split is invalid for forecasting. Randomly assigning rows lets the model train on Wednesday and Friday in order to predict Thursday — it learns to interpolate between known points, which is not the task. In production the model has only the past. Your evaluation must reproduce that constraint exactly.

CHRONOLOGICAL SPLIT



Validation approaches

APPROACH	HOW IT WORKS AND WHEN TO USE IT
Temporal hold-out	One chronological cut. Train on the earliest period, validate on the next, test on the most recent. Simple, fast, and the correct default.
Walk-forward validation	Train, predict the next block, roll the window forward, retrain, repeat. Produces several error estimates and shows whether accuracy is stable over time.
Time-series cross-validation	Expanding or sliding windows across multiple origins. Use when the series is long enough that a single split would be unrepresentative.

NON-NEGOTIABLE RULE

Future information must never reach the training process — not through row order, not through scalers fitted on the full dataset, not through target encodings computed across all time, and not through a validation set used repeatedly for tuning and then reported as a test result. Fit every transformation on training data only.

10 BASELINE FORECASTING MODEL

Build at least one simple baseline before touching an advanced model. A baseline converts an abstract score into a meaningful one: an MAE of 47 units means nothing on its own, but an MAE of 47 against a seasonal-naive baseline of 92 means your model has genuinely halved the error.

BASELINE	PREDICTION RULE	BEST SUITED TO
Naive	Tomorrow equals today	Series with strong short-term persistence and no seasonality
Seasonal naive	Next Monday equals last Monday	Any series with a clear weekly or annual cycle
Moving average	Mean of the last n observations	Noisy series with a stable level and weak seasonality

WHY BASELINES MATTER

If a gradient boosting model with forty engineered features cannot beat last week's value repeated forward, the problem is in your features, your split or your target — not in the model. The baseline is the fastest diagnostic you have.

11 FORECASTING MODELS

Experiment with several approaches from at least two families. You are not required to implement every model listed — you are required to select appropriate ones and justify the selection in writing.

FAMILY	MODEL	STRENGTHS	WATCH-OUTS
Statistical	ARIMA	Interpretable; strong on single, stable series	Needs stationarity; no native seasonality
Statistical	SARIMA	Handles seasonal structure explicitly	Parameter search is slow; struggles with multiple seasonalities
Statistical	Exponential smoothing	Fast, robust, excellent short-horizon baseline	Cannot use external regressors
Machine learning	Random Forest	Robust, minimal tuning, handles mixed features	Cannot extrapolate beyond the observed range
Machine learning	XGBoost	Very strong with engineered lag features	Requires careful tuning and leakage discipline
Machine learning	LightGBM	Fast on large multi-product panels	Sensitive to categorical handling
Machine learning	Gradient Boosting	Reliable general-purpose performer	Slower training than LightGBM
Advanced	Prophet	Holidays and changepoints handled natively	Can over-smooth sharp promotional spikes
Advanced	LSTM / GRU	Learns long dependencies without manual lags	Data-hungry; hard to justify on short series
Advanced	Temporal Fusion Transformer	Multi-horizon, multi-entity, attention-based	Heavy compute; genuinely optional

State clearly why you chose each model, and why you rejected the ones you did not use. A well-argued selection of three models is worth more than an unexplained implementation of eight.

12 MODEL TRAINING

Document the training process so that another person could reproduce your results from your repository alone.

STAGE	WHAT TO DO AND RECORD
Feature preparation	Assemble the feature matrix, sorted by time; drop rows where lags are undefined; fit scalers and encoders on training data only
Hyperparameter selection	Grid, random or Bayesian search evaluated on the validation period — never on the test period
Validation strategy	State whether you used a single hold-out or walk-forward validation, and the exact date boundaries
Training	Record library versions, random seeds, training duration and hardware
Prediction generation	Specify whether the forecast is direct multi-step or recursive, and how recursive lag values are supplied

Hyperparameters to document

MODEL	PARAMETERS RECORDED	SELECTED VALUE
LightGBM	n_estimators, learning_rate, num_leaves, max_depth	Record your final values
XGBoost	n_estimators, learning_rate, max_depth, subsample	Record your final values
SARIMA	(p, d, q) × (P, D, Q, s)	Record your final values
Prophet	seasonality_mode, changepoint_prior_scale, holidays	Record your final values

13 MODEL EVALUATION

Different metrics answer different questions, and a model that wins on one can lose on another. Report several, and explain which one you optimised for and why that choice suits the business decision at hand.

METRIC	WHAT IT MEASURES	WHEN TO PREFER IT
MAE	Average absolute error in original units	Default choice; easy to explain to non-technical stakeholders
RMSE	Root mean squared error; penalises large misses	When a single large miss is much worse than several small ones
MAPE	Average percentage error	Comparing across products of very different scale; fails on zero-demand days
sMAPE	Symmetric percentage error	Safer alternative to MAPE when the series contains zeros or near-zeros
R^2	Share of variance explained	Useful as a supporting figure; misleading on strongly trending series

Model comparison table

MODEL	MAE	RMSE	MAPE / SMAPE	TRAINING TIME
Seasonal naive (baseline)	—	—	—	—
ARIMA / SARIMA	—	—	—	—
Prophet	—	—	—	—

MODEL	MAE	RMSE	MAPE / SMAPE	TRAINING TIME
Random Forest	—	—	—	—
XGBoost / LightGBM	—	—	—	—

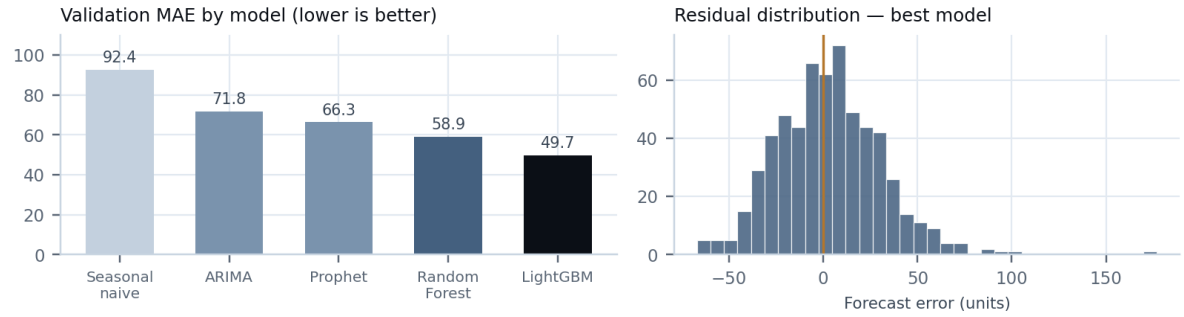


Figure 2 — Example evaluation view: validation error by model against the baseline, with the residual distribution of the selected model. Centred, roughly symmetric residuals indicate an unbiased forecast.

SELECT FOR THE DECISION, NOT THE LEADERBOARD

The final model should be chosen on predictive performance and practical considerations together: training and inference cost, retraining frequency, interpretability for the stakeholders who must trust it, and stability when the data shifts. A model two percent more accurate but impossible to explain or maintain is often the wrong answer.

14 FORECASTING RESULTS

Generate predictions over a horizon appropriate to your dataset and business use case. Longer horizons are not more impressive — they are simply less certain, and you must show that uncertainty rather than hide it.

HORIZON	TYPICAL USE	CONSIDERATIONS
7 days	Staffing and replenishment	Highest accuracy; weekly seasonality dominates
14 days	Short-cycle purchasing	Still reliable for most retail series
30 days	Monthly planning and procurement	Uncertainty widens noticeably; intervals become essential
90 days	Capacity and budget planning	Trend and season dominate; expect wide intervals

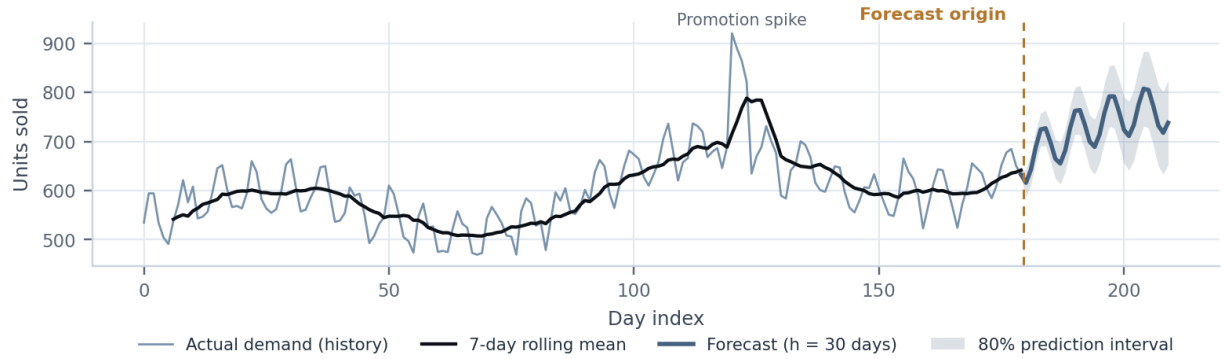


Figure 3 — Example forecast view: historical demand with a 7-day rolling mean, a 30-day forecast beyond the forecast origin, and an 80% prediction interval that widens with horizon.

What your results visualisation must show

- Actual demand plotted against predicted demand over the evaluation period
- A clearly marked forecast origin separating history from projection
- The forecast horizon, labelled with its length
- Confidence or prediction intervals wherever the model supports them
- High-demand periods the business should prepare for
- Periods of expected decline, so stock can be reduced in advance

> **SHOW THE UNCERTAINTY**
> A single forecast line implies a precision no model has. Where your method supports intervals, plot them. Where it does not, quote the historical error range alongside the point forecast so the reader understands the tolerance they are planning within.

15 PRODUCT & BUSINESS DEMAND INSIGHTS

This section converts model output into business meaning. Every statement here must trace back to a specific analytical finding — a chart, a metric or a forecast — and you should be able to point to it.

INSIGHT CATEGORY	WHAT TO REPORT
Growth products	Items with a rising forecast trajectory, quantified as expected change over the horizon
Declining products	Items with a falling trajectory and the point at which stock should be reduced
Seasonal peaks	Dates or weeks where demand is expected to exceed normal levels, and by how much
Inventory risk	Products where forecast demand exceeds current stock cover within lead time
Demand volatility	Products with wide prediction intervals that need larger safety stock
Low-demand items	Slow movers where holding cost is likely to exceed contribution

What should the business do with the forecast?

Translate each forecast into a decision. This table is the shape of the recommendation section your stakeholders will actually read.

FUNCTION	RECOMMENDED ACTION DRIVEN BY THE FORECAST
Inventory planning	Set reorder points and safety stock from forecast demand and its uncertainty over supplier lead time
Procurement	Schedule purchase orders ahead of predicted peaks; defer orders for declining lines
Production	Align production runs to the forecast profile to reduce changeovers and idle capacity
Staffing	Roster to predicted daily volume; pre-plan cover for the highest forecast days
Marketing	Promote where demand is soft and margin allows; avoid discounting items already forecast to peak
Sales planning	Set targets from the forecast range rather than last year plus an arbitrary uplift
Supply chain	Reallocate stock towards locations with the strongest forecast growth

> **EVIDENCE, NOT OPINION**
> Recommendations must be grounded in your actual analytical findings. Every claim should be traceable to a chart, a metric or a forecast in your repository. Unsupported advice is the fastest way to lose the trust the forecast was built to earn.

16 INTERACTIVE FORECASTING DASHBOARD

Build an interactive dashboard using Streamlit, Power BI, Tableau, Plotly Dash or another suitable framework. The dashboard is how a non-technical stakeholder experiences your work — it should be understandable without you standing next to it.

PANEL	REQUIRED CONTENT
Overview	Total historical sales, average demand, total forecasted demand, and a clear growth or decline indicator
Demand trends	Historical demand, forecasted demand and identified seasonal patterns on a shared timeline
Product analysis	Product selector, product-level forecast and a ranked view of top-demand products
Forecast performance	Model comparison with MAE, RMSE and MAPE or sMAPE, including the baseline
Business insights	Expected demand increases, expected declines and flagged inventory risks

Interface standards

- One clear message per panel; if a chart needs a paragraph to explain, redesign the chart
- Consistent colour meaning throughout — for example one colour for actuals and one for forecast, everywhere
- Readable defaults: the dashboard must be useful before the user touches a single filter
- Units and dates labelled on every axis; no raw column names visible in the interface
- Loading behaviour handled gracefully — cache computation rather than recomputing on every interaction

17 OPTIONAL & ADVANCED FEATURES

The core scope above is sufficient for a complete submission. The following extensions are optional and are credited where the core work is already strong.

ANALYTICS EXTENSIONS	ENGINEERING EXTENSIONS	RESEARCH EXTENSIONS
Multi-product forecasting	Real-time forecasting API	LSTM / GRU forecasting
Multi-store forecasting	FastAPI backend service	Transformer-based forecasting
Forecast confidence intervals	Automated retraining pipeline	Automated model selection
What-if scenario analysis	Model performance monitoring	Hyperparameter optimisation
Promotion impact analysis	Forecast drift detection	Hierarchical reconciliation
Price–demand analysis	Automated low-stock alerts	Probabilistic / quantile forecasting
Inventory recommendation engine	Scheduled reporting	Ensembling across model families

18 ERROR & FORECAST ANALYSIS

Analysing where your system fails is one of the strongest signals of professional maturity in this project. A candidate who reports a clean error analysis is more trusted than one who reports only a good score.

WHERE IT FAILS	WHY IT FAILS	WHAT WOULD HELP
Demand spikes	The driver was never in the training data	Promotion calendar, campaign spend, event data
Sudden drops	Stockouts recorded as zero demand rather than unmet demand	Inventory levels and availability flags
Season changes	Too few complete cycles for the model to learn the pattern	Longer history or an explicit seasonal term
Promotional periods	Uplift is non-linear and depends on discount depth	Discount percentage and promotion type
Missing data	Gaps break lag features and distort rolling windows	A documented imputation policy and gap flags
Unusual events	Weather, closures, competitor action are outside the model	Weather data, holiday calendar, competitor pricing

Three questions to answer explicitly

- Where does the model fail? Identify the specific dates, products or conditions with the largest errors.
- Why might it fail there? Attribute the error to a cause: missing driver, insufficient history, data quality or genuine unpredictability.
- What additional data would improve the forecast? Name the source, the expected effect and how you would validate the improvement.

> **LOOK AT RESIDUALS OVER TIME, NOT JUST IN AGGREGATE**
> Plot forecast error against date. Errors that drift, widen or cluster around particular weekdays or promotions reveal structure the model has not captured — and each pattern points directly at a feature you have not yet built.

19 BUSINESS SCENARIO

Choose one realistic business scenario and commit to it. The entire project — dataset, horizon, metric and recommendations — should follow from that choice.

INDUSTRY	BUSINESS PROBLEM	TYPICAL TARGET AND HORIZON
Retail	Predict product demand per store	Units sold per SKU per day; 7–30 days
E-commerce	Forecast future order volume	Orders per day; 14–30 days
Restaurant	Predict daily meal demand	Covers or dishes per day; 7–14 days
Manufacturing	Forecast production requirements	Units per production cycle; 30–90 days
Energy	Predict future energy consumption	kWh per hour or day; 1–30 days
Logistics	Forecast shipment or order volumes	Shipments per day; 7–30 days

> **STATE THE CHAIN EXPLICITLY IN YOUR README**
> Industry → Business problem → Dataset → Forecasting approach → Prediction → Business decision. Six links, one line each. If any link is missing, the project reads as a modelling exercise rather than a solution.

20 EXPECTED DELIVERABLES

Every item below must be present in your submission. Anything marked as optional in earlier sections is excluded from this list.

☐ Dataset and/or reproducible data pipeline	☐ Documented data cleaning workflow
☐ Data dictionary for every field used	☐ Exploratory data analysis with interpretation
☐ Time-series analysis and decomposition	☐ Feature engineering with stated rationale
☐ Baseline forecasting model	☐ Multiple forecasting models compared
☐ Model evaluation with several metrics	☐ Forecast visualisations
☐ Error and failure analysis	☐ Business insights and recommendations
☐ Interactive forecasting dashboard	☐ GitHub repository, well organised
☐ Professional README.md	☐ Screenshots of dashboard and key charts
☐ LinkedIn project demonstration video	

21 GITHUB REQUIREMENTS

Your repository is the primary artefact a reviewer or employer will open. Structure it so that someone can understand the project in two minutes and reproduce it in ten.

```
project/
│
├── data/            raw and processed data, or a download script
├── notebooks/       EDA, decomposition, modelling experiments
├── src/             reusable code: features, training, evaluation
├── models/          serialised models and configuration
├── dashboard/       dashboard application source
├── reports/         findings, metrics, business recommendations
├── images/          exported charts and dashboard screenshots
├── requirements.txt
├── README.md
└── .gitignore
```

Required README.md sections

#	SECTION	#	SECTION
01	Project Overview	09	Model Evaluation

#	SECTION	#	SECTION
02	Business Problem	10	Forecast Results
03	Dataset	11	Dashboard
04	Data Dictionary	12	Business Insights
05	EDA	13	Installation
06	Time-Series Analysis	14	Usage
07	Feature Engineering	15	Limitations
08	Forecasting Models	16	Future Improvements

SECURITY

Never upload API keys, passwords, credentials, private data or any sensitive information to GitHub. Use environment variables and a .env file excluded by .gitignore. Check your notebook outputs as well as your source files — credentials are most often leaked in a cell that was never cleared.

22 LINKEDIN DEMONSTRATION

Record a short demonstration video and publish it with a written post. This is how the work becomes visible to people who will never open your repository.

#	SEGMENT	SUGGESTED LENGTH	WHAT TO SHOW
01	Dataset overview	15–20 s	Source, size, time period, target variable
02	EDA visualisations	20–30 s	Trend and seasonality, with one sentence of interpretation each
03	Forecasting model	20–30 s	Models tried and why the final one was selected
04	Actual vs predicted	20–30 s	Evaluation chart and headline accuracy against the baseline
05	Future forecast	15–20 s	The forward projection with its uncertainty interval
06	Interactive dashboard	30–45 s	A live walkthrough, not a static screenshot
07	Business insights	20–30 s	Two or three concrete recommendations

What the written post should cover

- The business problem in one clear sentence, free of technical vocabulary
- The dataset and why it suited the problem
- Your data science approach, from cleaning through to feature engineering
- Models used and the reasoning behind the final selection
- Evaluation results, stated against the baseline rather than in isolation
- The dashboard and who it was designed for
- Key findings and the decisions they support
- Lessons learned — including what did not work and what you would do differently

23 SUBMISSION REQUIREMENTS

Complete every item below before submitting. Incomplete submissions are returned rather than assessed.

- ☐ GitHub repository link
- ☐ LinkedIn project demonstration link
- ☐ Dashboard link or screenshots where applicable
- ☐ Full name exactly as it should appear on your certificate
- ☐ Required email and contact information
- ☐ Any additional information requested through the official seKer AI submission process

24 PROJECT QUALITY STANDARDS

Your submission is assessed across fourteen criteria. Strength in modelling does not compensate for weakness in analysis, documentation or communication.

CRITERION	WHAT A STRONG SUBMISSION LOOKS LIKE
Data quality	Cleaning decisions are deliberate, documented and defensible
EDA depth	Analysis goes beyond descriptive statistics into diagnosis
Time-series understanding	Trend, seasonality and stationarity are addressed explicitly
Feature engineering	Features encode stated hypotheses rather than being generated blindly
Model selection	Choices are justified against the problem and the data
Forecasting methodology	Horizon, granularity and prediction strategy are coherent
Leakage prevention	Chronological integrity is demonstrably maintained throughout
Evaluation quality	Multiple metrics, compared against a baseline, honestly reported
Visualisation quality	Charts are labelled, purposeful and interpreted
Dashboard usability	Understandable without narration by the author
Business insight	Recommendations are specific, actionable and evidence-backed
Code quality	Readable, modular, reproducible, sensibly named
GitHub documentation	A reviewer can run the project from the README alone
Overall presentation	The work reads as a professional deliverable end to end

> **THE OBJECTIVE**
> The objective is not simply to produce a forecast. The objective is to build a reliable predictive analytics solution that helps a real business make better decisions.

25 LEARNING OUTCOMES

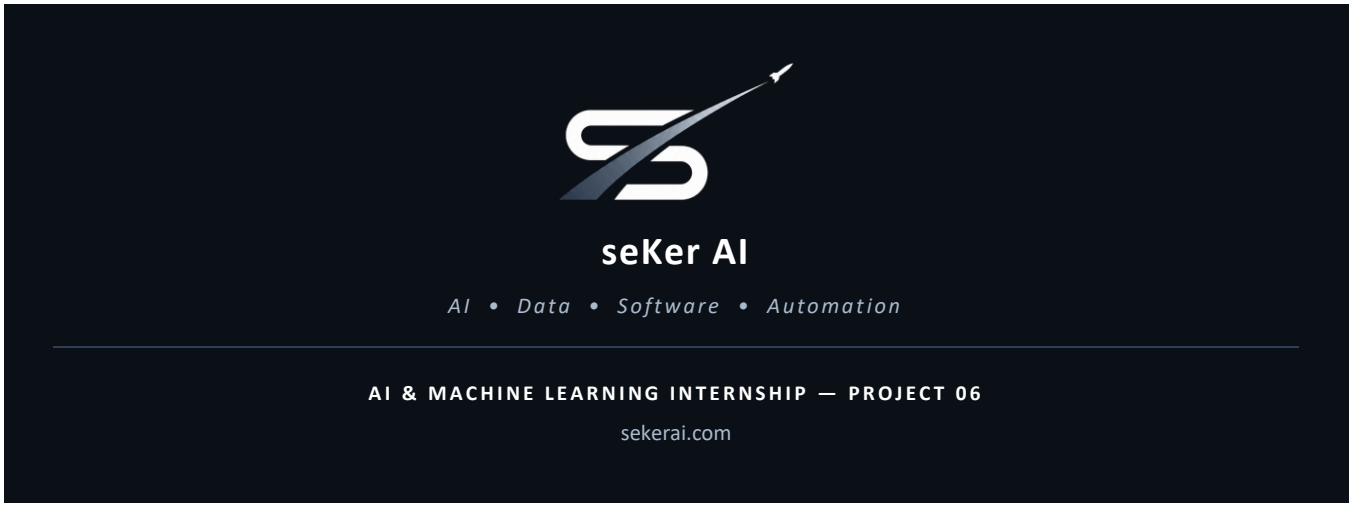
On completing Project 06 you will have practical, demonstrable experience across the full applied forecasting stack — the same sequence of skills used in commercial demand planning teams.

ANALYTICAL & MODELLING SKILLS	DELIVERY & PROFESSIONAL SKILLS
Time-series analysis	Time-series validation
Data preprocessing	Avoiding data leakage
Exploratory data analysis	Predictive analytics
Feature engineering	Data visualisation
Forecasting methodology	Dashboard development
Statistical models	Business decision-making
Machine learning	GitHub documentation
Model evaluation	Professional project presentation

26 FINAL PROJECT CHECKLIST

Work through this checklist before you submit. Each item corresponds to a section of this document.

☐ Dataset selected and licence confirmed	☐ Data dictionary created
☐ Data cleaned and decisions documented	☐ EDA completed with interpretation
☐ Trends identified	☐ Seasonality analysed
☐ Time-series decomposition performed	☐ Features engineered and justified
☐ Chronological train / test split implemented	☐ Baseline model developed
☐ Multiple forecasting models tested	☐ Models evaluated on several metrics
☐ Best model selected and justified	☐ Future demand forecast generated
☐ Error analysis completed	☐ Business insights generated
☐ Dashboard created and tested	☐ GitHub repository completed
☐ README.md completed	☐ LinkedIn demo published
☐ Final submission completed	


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